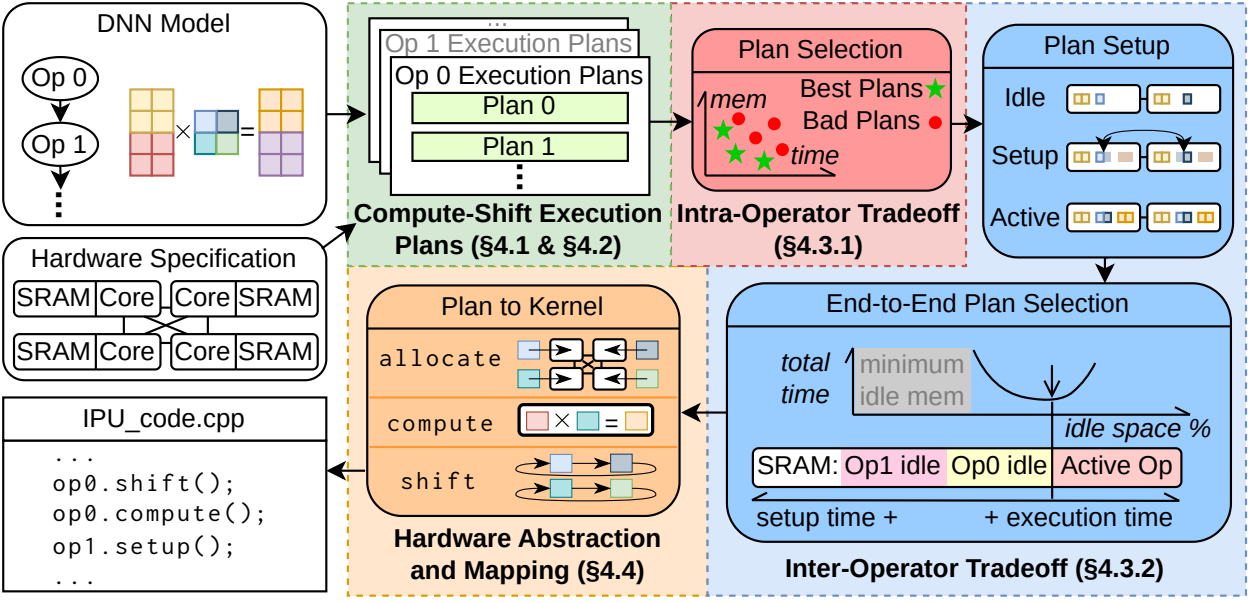

DNN Model
Op 0
Op 1
Hardware Specification
SRAM
Core
Core
SRAM
SRAM
Core
Core
SRAM
IPU_code.cpp
...
op0.shift();
op0.compute();
op1.setup();
...
...
Op 1 Execution Plans
Op 0 Execution Plans
Plan 0
Plan 1
Compute-Shift Execution Plans (§4.1 & §4.2)
Plan Selection
mem
Best Plans
Bad Plans
time
Intra-Operator Tradeoff (§4.3.1)
Plan Setup
Idle
Setup
Active
Plan to Kernel
allocate
compute
shift
Hardware Abstraction and Mapping (§4.4)
End-to-End Plan Selection
total time
minimum idle mem
idle space %
SRAM: Op1 idle
Op0 idle
Active Op
setup time +
+ execution time
Inter-Operator Tradeoff (§4.3.2)